

Parvaneh Janbakhshi

*Sr. Data Scientist at Bayer AG
Berlin, Germany*

✉ pr.janbakhshi@gmail.com
📁 pjanbakhshi.github.io

Interests

AI Solutions for Healthcare and Life Sciences, Speech and Audio Signal Processing for Clinical Applications, Machine Learning and Deep Learning, Medical Image Analysis, Biomedical Signal Processing

Education

- 2018–2022 **Doctor of Philosophy (PhD) in Electrical Engineering**, *École Polytechnique Fédérale de Lausanne (EPFL)*, Lausanne, Switzerland.
GPA: 5.5/6, 12 credits
- 2014–2016 **Master of Science in Biomedical Engineering (Bioelectronics)**, *Sharif University of Technology*, Tehran, Iran.
GPA: 4/4 (18.88/20), 29 credits – ranked 2nd
- 2009–2014 **Bachelor of Science in Biomedical Engineering (Bioelectronics)**, *Amirkabir University of Technology (Tehran Polytechnic)*, Tehran, Iran.
GPA: 3.70/4 (17.27/20), 140 credits

Research & Work Experience

- 2025–Present **Senior Data Scientist** at Bayer AG, Berlin, Germany
- 2022–2025 **Data Scientist** at Bayer AG, Berlin, Germany
- Projects: AI-powered experiment documentation for pharmaceutical chemistry, manufacturing & controls (CMC), Detection of adverse events (AEs) from textual data, MRI contrast enhancement, Parkinson's disease vocal biomarkers
- 2022 **Postdoctoral Researcher** at Idiap Research Institute, Martigny, Switzerland
- 2018–2022 **Doctoral Researcher** at Idiap Research Institute, Martigny, Switzerland
- Thesis: Automatic pathological speech assessment
 - Selected coursework: Deep Learning, Optimization for Machine Learning, Statistical Sequence Processing
- 2017 **Intern** at Cognitive Neurobiology Laboratory in Cognitive Science Institute, Tehran, Iran
- Project: Investigating the phase-amplitude coupling in the middle temporal visual area of Rhesus monkeys
- 2016 **Master of Science Thesis** at Sharif University of Technology, Tehran, Iran
- Thesis: Extraction of respiratory information from ECG and its application for sleep apnea detection
 - Selected coursework: Pattern Recognition, Biological Signal Processing
- 2014 **Bachelor of Science Thesis** at Amirkabir University of Technology, Tehran, Iran

- Thesis: Designing and implementing an automatic neuromuscular electro-stimulation device to prevent deep vein thrombosis and varicose veins (*patented in Iran, No. 83492*)

Selected Publications

- For a complete list of publications and patents, please visit my Google Scholar Profile.
- Conference I. M. Baltruschat, **P. Janbakhshi**, M. Dohmen and M. Lenga, "Uncertainty estimation in contrast-enhanced MR image translation with multi-axis fusion", In Proc. ISBI, 2024.
- Conference **P. Janbakhshi**, I. Kodrasi, "Adversarial-free speaker identity-invariant representation learning for automatic dysarthric speech classification", In Proc. INTERSPEECH, 2022.
- Conference **P. Janbakhshi**, I. Kodrasi, "Supervised speech representation learning for Parkinson's disease classification", In Proc. ITG Conference on Speech Communication, 2021.
- Journal **P. Janbakhshi**, I. Kodrasi, H. Bourlard, "Subspace-based learning for automatic dysarthric speech detection", Signal Processing Letters, 2021
- Journal **P. Janbakhshi**, I. Kodrasi, H. Bourlard, "Automatic pathological speech intelligibility assessment exploiting subspace-based analyses," IEEE/ACM Transactions on Audio, Speech, and Language Processing, 2020
- Conference **P. Janbakhshi**, I. Kodrasi, H. Bourlard, "Pathological speech intelligibility assessment based on the short-time objective intelligibility measure", In Proc. ICASSP, 2019
- Journal **P. Janbakhshi**, M. B. Shamsollahi, "Sleep apnea detection from single-lead ECG using features based on ECG-derived respiration (EDR) signals", IRBM, 2018.

Honors & Awards

- 2023 Individual performance award (IEZ), Bayer AG
- 2021 PhD student award by Idiap Research Institute
- 2016 Ranked 2nd in the Master of Science Bioelectric program at Sharif University of Technology
- 2014 Ranked 50th among more than 15000 participants in nationwide university entrance exam in MSc for Biomedical Engineering
- 2013 Best BSc project of the year in Bioelectric Engineering department

Academic Duties & Mentorships

- 2026-present **Development Coach** at DSAIF team, Bayer AG
- 2023-2025 **Intern Supervisor** at DSAIF team, Bayer AG
 - Projects: AI for Parkinson's disease voice assessment, AI for cardiac biomarker extraction
- 2021 **Teaching Assistant** for Linear Algebra and Probability course of Master in Artificial Intelligence, UniDistance Suisse
- 2020–present **Reviewer for Journals:** Nature's Scientific Reports, Springer's Behavior Research Methods, Elsevier's Computers in Biology and Medicine, Elsevier's Speech Communication, IEEE/ACM Transactions on Audio, Speech, and Language Processing
- 2020–present **Reviewer for Conferences:** IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), Annual Conference of the International Speech Communication Association (Interspeech)

Skills

Programming Python

Cloud & DevOps	AWS, SageMaker, Terraform, CI/CD, GitHub Actions
Machine Learning & AI	PyTorch
Software & Frameworks	PyTest, Hydra, SimpleITK, PyDICOM, MATLAB, Praat, L ^A T _E X
Languages	English (fluent), German (C1, Goethe-Zertifikat), Farsi (native)